

HOMEWORK #1

1. **[20 pts] [Basic Concept of “Electromagnetics]** Please succinctly answer for the following questions:
 - (a) [4 pts] “What is “Electromagnetics”?”
 - (b) [4 pts] What are the four fundamental SI unit in the electromagnetics?
 - (c) [6 pts] What are the four fundamental field quantities in the electromagnetics model? and write their units
 - (d) [6 pts] What are the three-universal constants in the electromagnetic model? and find the permittivity in free space, ϵ_0 , of material from the physical relationship between c and magnetic permeability in free space, μ_0

2. **[20 pts] [Understanding of Vector Presentation in terms of Orthogonal Rectangular Components]** In rectangular coordinates, given points $M(-1, 2, 1)$, $N(3, -3, 0)$, and $P(-2, -3, -4)$. Find:
 - (a) [5 pts] $\overrightarrow{A_{MN}}$, Magnitude of $\overrightarrow{A_{MN}}$, and Direction of Unit Vector, $\widehat{a_{MN}}$
 - (b) [5 pts] $\overrightarrow{A_{MN}} + \overrightarrow{A_{MP}}$
 - (c) [5 pts] $|\overrightarrow{A_M}|$, and $|\overrightarrow{A_N}|$
 - (d) [5 pts] $|5\overrightarrow{A_p} - 2\overrightarrow{A_N}|(\overrightarrow{A_{MP}} + \overrightarrow{A_{PN}})$

3. **[30 pts] [Understanding of Vector Field and Dot and Cross Products]** In Cartesian coordinates, vector $\vec{A}$ is directed from the origin to point $P_1(2, 3, 3)$, and vector $\vec{B}$ is directed from P_1 to point $P_2(1, -2, 2)$. Find:
 - (a) [3 pts] vector $\vec{A}$, its magnitude A , and unit vector $\hat{a}$,
 - (b) [4 pts] the angle that $\vec{A}$ makes with the y -axis,
 - (c) [3 pts] vector $\vec{B}$
 - (d) [4 pts] the angle between $\vec{A}$ and $\vec{B}$
 - (e) [4 pts] the perpendicular distance from the origin to vector $\vec{B}$
 - (f) [6 pts] $\vec{A} \cdot \vec{B}$, and $\vec{B} \cdot \vec{A}$. Compare the calculation results are the same or not.
 - (g) [6 pts] $\vec{A} \times \vec{B}$ and $\vec{B} \times \vec{A}$

4. **[10 pts] [Understanding of Basic Law of Cross Vector Products]** Given $\vec{A} = \hat{x} - \hat{y} + \hat{z}2$, $\vec{B} = \hat{y} + \hat{z}$, and $\vec{C} = -\hat{x}2 + \hat{z}3$. Find $(\vec{A} \times \vec{B}) \times \vec{C}$ and compare with $\vec{A} \times (\vec{B} \times \vec{C})$.

5. [20 pts] **(HARD PROBLEM)** [Understanding of Two Vector Positions Based on the Geometrical Properties of Dot and Cross Products] Given $\vec{A} = \hat{a}_x 5 - \hat{a}_y 2 + \hat{a}_z$, find the expression of a unit vector $\vec{B}$ in the case under below situations
- (a) $\vec{B} \parallel \vec{A}$
- (b) $\vec{B} \perp \vec{A}$, if $\vec{B}$ lies in the xy -plane